

A critical Allen–Cahn energy counterexample

The expected squared path supremum is infinite at $\gamma^2 = 2$

Automated mathematics run `fa_banach_001`
Lane 6 review packet

21 July 2026

Abstract

Problem 4 of Agresti–Veraar asks for a critical L^2 energy-supremum estimate and highlights the periodic Allen–Cahn equation with quadratic noise at $\gamma^2 = 2$ as unknown. We give a negative answer. For every positive deterministic constant initial condition, the spatially constant solution is the reciprocal radius of a four-dimensional Ornstein–Uhlenbeck process. A second-moment estimate for its small-ball occupation time yields a $1/y$ lower tail for the squared reciprocal-radius supremum. Thus the supremum is finite almost surely on every compact time interval, but its expectation is infinite.

Packet status. Candidate counterexample, likely valid; prepared for expert review. The probabilistic argument is self-contained apart from the standard fact that points are polar for four-dimensional Brownian motion, for which a one-line annulus argument is recalled below.

1 Source question

Agresti and Veraar ask whether, in the critical variational setting, one has

$$\mathbb{E} \sup_{t \in [0, T]} \|u(t)\|_H^2 \leq C_T (1 + \mathbb{E} \|u_0\|_H^2). \quad (1)$$

They then single out the one-dimensional periodic equation

$$du = (\Delta u + u - u^3) dt + \gamma u^2 dW \quad (2)$$

and state that the L^2 bound is unknown at $\gamma^2 = 2$ [1, Problem 4, p. 92]. The following image is a direct crop from the current source PDF.

2 Result

Lebesgue measure on $\mathbb{T}$ may have any fixed positive normalization, denoted by $|\mathbb{T}|$.

Theorem 1 (Negative answer at the critical coefficient). *Fix $x_0 > 0$, set $\gamma = \sqrt{2}$, and take the smooth deterministic initial condition $u_0(\xi) = x_0$ in (2). For every $T > 0$, the resulting global spatially constant solution satisfies*

$$\mathbb{E} \sup_{0 \leq t \leq T} \|u(t)\|_{L^2(\mathbb{T})}^2 = \infty.$$

In particular, no finite constant C_T can make (1) hold for this critical model.

Proof intuition

Constant functions form an invariant mode, so the SPDE contains a scalar SDE. For a general positive γ , taking the reciprocal and rescaling formally produces a radial Ornstein–Uhlenbeck process of dimension

$$\delta = 3 + \frac{2}{\gamma^2}.$$

*L*²-moments in the variational setting. In Theorem 6.4 we obtained a bound for $\mathbf{E} \sup_{t \in [0, T]} \|u(t)\|_H^p$ for $p \in (0, 2)$, but $p = 2$ is excluded in general. In Section 6.3.1, conditions are discussed under which one can extend this bound to $p = 2$. However, we do not know whether this condition can be omitted or not.

Problem 4 (Energy bounds for critical variational SPDEs). *In the variational setting of Theorem 6.4, is there a constant C_T such that for all $u_0 \in L^2_{\mathcal{F}_0}(\Omega; H)$ one has*

$$\mathbf{E} \sup_{t \in [0, T]} \|u(t)\|_H^2 \leq C_T (1 + \mathbf{E} \|u_0\|_H^2)?$$

A model example for which we do not know whether the L^2 -bound holds is the Allen–Cahn equation on the one-dimensional torus $\mathbb{T}$ with a one-dimensional quadratic noise:

$$du = (\Delta u + u - u^3) dt + \gamma u^2 dW.$$

For $\gamma^2 < 2$, Section 6.3.1 gives an $L^2(\Omega)$ -bound. But for $\gamma^2 = 2$ we do not know whether it holds.

In Theorem 6.4 we stated a result on the continuous dependency on the initial data in terms of convergence in probability. In [AV24, Theorem 3.8] it is also explained that the following holds as well: if $u_0^n \rightarrow u_0$ in $L^2(\Omega; H)$, then $u^n \rightarrow u$ in $L^p(\Omega; L^2(0, T; V) \cap C([0, T]; H))$ for any $p \in (0, 2)$. It is natural to ask when convergence takes place for $p = 2$.

Figure 1: Problem 4 and the critical Allen–Cahn model in arXiv:2501.18561v5, page 92.

The critical value $\gamma^2 = 2$ is therefore exactly dimension four. A four-dimensional diffusion visits an ε -ball during a fixed time window with probability of order at least ε^2 . Since the original solution is the reciprocal radius, such a visit creates a squared peak of order ε^{-2} . The two powers cancel, producing the borderline $1/y$ tail whose integral diverges. The formal proof below derives the four-dimensional representation and the required lower bound without assuming independence of visits.

3 Reciprocal radial representation

On the spatially constant subspace, (2) becomes

$$dX_t = (X_t - X_t^3) dt + \sqrt{2} X_t^2 dW_t, \quad X_0 = x_0. \quad (3)$$

For as long as $X > 0$, Itô's formula with $Y = X^{-1}$ gives

$$\begin{aligned} dY_t &= -X_t^{-2} dX_t + X_t^{-3} d[X]_t \\ &= \left(-Y_t + \frac{3}{Y_t}\right) dt - \sqrt{2} dW_t. \end{aligned}$$

Consequently $R = Y/\sqrt{2}$ obeys

$$dR_t = \left(-R_t + \frac{3}{2R_t}\right) dt + d\beta_t, \quad \beta = -W. \quad (4)$$

We can construct this process globally. Let Z be the four-dimensional OU process

$$dZ_t = -Z_t dt + dB_t^{(4)}, \quad |Z_0| = \frac{1}{\sqrt{2}x_0}. \quad (5)$$

Itô's formula for the Euclidean norm shows that $R_t = |Z_t|$ satisfies (4), with $\beta_t = \int_0^t \langle Z_s / |Z_s|, dB_s^{(4)} \rangle$ up to a possible zero-hitting time. There is no such time: explicitly,

$$e^t Z_t = Z_0 + \tilde{B}_{a(t)}, \quad a(t) = \frac{e^{2t} - 1}{2},$$

for a four-dimensional Brownian motion $\tilde{B}$. A Brownian motion started at radius $r > 0$ hits the ball of radius $\varepsilon < r$ with probability $(\varepsilon/r)^2$; this follows by optional stopping of the harmonic function $|z|^{-2}$ on annuli. Letting $\varepsilon \downarrow 0$ shows that a prescribed point is polar. Hence $R_t > 0$ for every t almost surely.

Now set

$$X_t = \frac{1}{\sqrt{2}|Z_t|}, \quad W = -\beta. \quad (6)$$

A second application of Itô's formula verifies (3). This is a positive global solution. Moreover, continuity and nonvanishing of Z imply that $\min_{0 \leq t \leq T} |Z_t| > 0$ almost surely, so its pathwise supremum is finite on every compact interval. Under the well-posedness asserted for the source model, uniqueness identifies this solution with the variational solution for the constant initial datum.

4 A fixed-window small-ball estimate

The next lemma is the quantitative core of the counterexample.

Lemma 1. *Let Z solve (5). For every $T > 0$ there are constants $c > 0$ and $\varepsilon_0 > 0$ such that*

$$\mathbb{P}\left(\inf_{T/2 \leq t \leq T} |Z_t| \leq \varepsilon\right) \geq c\varepsilon^2, \quad 0 < \varepsilon \leq \varepsilon_0.$$

Proof. Put $I = [T/2, T]$ and

$$O_\varepsilon = \int_I \mathbf{1}_{\{|Z_t| \leq \varepsilon\}} dt.$$

For $t > 0$, Z_t is Gaussian with mean $e^{-t}Z_0$ and covariance

$$q_t I_4, \quad q_t = \frac{1 - e^{-2t}}{2}.$$

As t ranges over the compact interval I , these densities are uniformly bounded above on $\mathbb{R}^4$ and uniformly bounded below on the unit ball. Thus, for $0 < \varepsilon \leq 1$,

$$c_1 \varepsilon^4 \leq \mathbb{P}(|Z_t| \leq \varepsilon) \leq C_1 \varepsilon^4, \quad t \in I. \quad (7)$$

In particular, $\mathbb{E}O_\varepsilon \geq c_2 \varepsilon^4$.

If $s < t$ are in I and $h = t - s$, then conditionally on Z_s ,

$$Z_t = e^{-h}Z_s + G_h, \quad G_h \sim N(0, q_h I_4).$$

On $0 < h \leq T/2$, one has $q_h \geq c_T h$. Bounding a Gaussian probability by the volume of the ball times the maximum of its density gives, uniformly in the conditional mean,

$$\mathbb{P}(|Z_t| \leq \varepsilon \mid Z_s) \leq \min\{1, C_2 \varepsilon^4 h^{-2}\}. \quad (8)$$

Combining (7) and (8), then integrating in s and h , yields

$$\begin{aligned} \mathbb{E}O_\varepsilon^2 &\leq C_3 \varepsilon^4 \int_0^{T/2} \min\{1, C_2 \varepsilon^4 h^{-2}\} dh \\ &\leq C_4 \varepsilon^6. \end{aligned}$$

For the last inequality, split the integral at $h = \sqrt{C_2} \varepsilon^2$ (and decrease ε_0 if needed). Cauchy–Schwarz on $O_\varepsilon \mathbf{1}_{\{O_\varepsilon > 0\}}$ now gives

$$\mathbb{P}(O_\varepsilon > 0) \geq \frac{(\mathbb{E}O_\varepsilon)^2}{\mathbb{E}O_\varepsilon^2} \geq c_3 \varepsilon^2.$$

The event $\{O_\varepsilon > 0\}$ is contained in the hitting event stated in the lemma. $\square$

5 Proof of Theorem 1

Let $M_T = \sup_{0 \leq t \leq T} X_t^2$. By (6) and Lemma 1,

$$\mathbb{P}\left(M_T \geq \frac{1}{2\varepsilon^2}\right) \geq c\varepsilon^2 \quad (0 < \varepsilon \leq \varepsilon_0).$$

With $\varepsilon = (2y)^{-1/2}$, this becomes

$$\mathbb{P}(M_T \geq y) \geq \frac{c}{2y}$$

for all sufficiently large y . The layer-cake formula therefore gives

$$\mathbb{E}M_T = \int_0^\infty \mathbb{P}(M_T \geq y) dy = \infty.$$

Finally, the constant solution $u(t, \xi) = X_t$ satisfies

$$\|u(t)\|_{L^2(\mathbb{T})}^2 = |\mathbb{T}|X_t^2,$$

whereas $\mathbb{E}\|u_0\|_{L^2(\mathbb{T})}^2 = |\mathbb{T}|x_0^2 < \infty$. This contradicts (1) for every finite C_T and completes the proof.

6 Checks, compatibility, and scope

- The divergence concerns the expectation of the *path supremum*; it does not assert finite-time explosion. Indeed, every sample path is bounded on $[0, T]$.
- At each fixed $t > 0$, $\mathbb{E}X_t^2 < \infty$: a nondegenerate density in four dimensions is locally integrable against $|z|^{-2}$. Thus the result is compatible with ordinary fixed-time energy moments.
- The verifier script in `code/verify_reduction.py` checks the three exact Itô/generator identities and the elementary split-integral scaling. It is a guard against algebraic errors, not a substitute for Lemma 1.
- The theorem resolves the highlighted $\gamma^2 = 2$ model negatively and therefore refutes Problem 4 as a general assertion. It says nothing about the neighboring continuous-dependence Problem 5 or the subcritical regime $\gamma^2 < 2$.

Literature-status note

A bounded search on 21 July 2026 checked the run indexes, the current arXiv record, exact phrases from Problem 4, and searches combining the critical quadratic-noise model with reciprocal and radial-OU transforms. No later proof or counterexample to this exact question was found. This statement records the search performed; it is not a claim of exhaustive priority.

References

- [1] Antonio Agresti and Mark Veraar, *Nonlinear SPDEs and maximal regularity: an extended survey*, arXiv:2501.18561v5, 2025.